Lab 9b: Magnetic Force on a Wire

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Pre-lab Questions

One consequence of placing a current carrying wire in a magnetic field is that the wire experiences a force $\vec{F} = I\vec{L} \times \vec{B}$ on it, where I is the current, L is the length of the wire within the magnetic field, and $\vec{B}$ is the magnetic field vector.

Note the use of the cross product, which both modifies the magnitude of the force, and defines its direction. If I, L, and B are expressed in units of Amps, meters, Teslas respectively, then the force is expressed in Newtons.

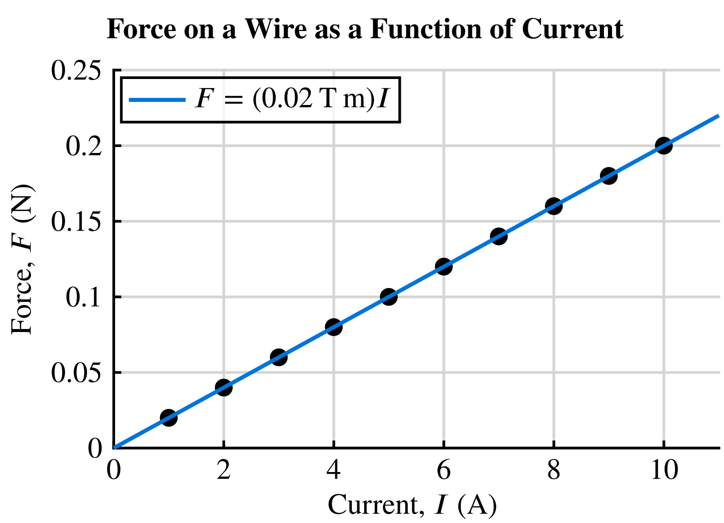
1. Make a properly labeled plot of the Force on a wire of length of wire 1 cm long that is perpendicular to a magnetic field of 2 T, as the current ranges from 1 A to 10 A. What is the slope of this plot? What does that slope represent?

   Since the wire is perpendicular to the field, $\sin\theta = 1$, and the force simplifies to:

   $$F = ILB$$

   With $L = 1\text{ cm} = 0.01\text{ m}$ and $B = 2\text{ T}$:

   $$F = I \cdot 0.01\text{ m} \cdot 2\text{ T} = (0.02\text{ T m})I$$

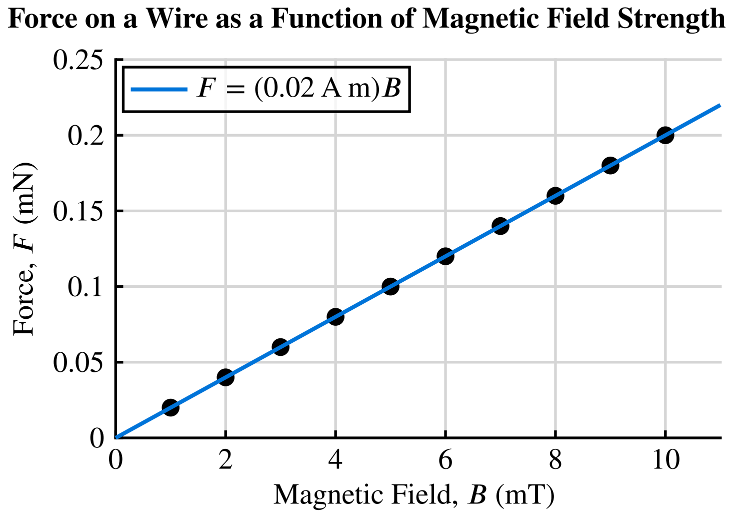
   

   The slope of this plot is 0.02 T m. It represents the product LB, i.e. the length of the wire multiplied by the magnetic field strength.

2. Make a properly labeled plot of the Force on a wire of length of wire 1 cm long with a current of 2 A. The wire is perpendicular to a magnetic field that ranges from 1 mT to 10 mT. What is the slope of this plot? What does that slope represent?

   With $L = 0.01\text{ m}$ and $I = 2\text{ A}$:

   $$F = ILB = 2\text{ A} \cdot 0.01\text{ m} \cdot B = (0.02\text{ A m})B$$



The slope of this plot is 0.02 A m. This slope represents the product $I\,L$, i.e. the current through the wire multiplied by the length of the wire.

Abstract

In this lab, we attempted to determine the magnetic field inside the opening of a permanent magnet by measuring the vertical magnetic force exerted by the magnetic field on a current-carrying wire. The magnitude of this magnetic force is given by $F = ILB$, where I is the current flowing through the wire, L is the length of the wire, and B is the magnetic field strength. So, by fixing I or L and measuring F, we can extract the magnetic field strength B. This verifies the linear dependence of I and L on F.

For each trial, we lowered the current-carying wire into the opening of the permanent magnet. The scale was rebalanced, and a mass measurement was taken. The change in mass was recorded to calculate the change in vertical force (treated as the difference from gravitational force), which we assumed to be the magnetic force F. Over multiple measurements of F, we performed a linear regression to extract the magnetic field strength B. This procedure was performed twice: once where current was fixed and once where wire length was fixed.

Finally, we directly measured the magnetic field strength using a magnetic field sensor, and compared this value to our previous two estimates of B. With I fixed, the estimate had a percent difference of 2.9% from the sensor measurement. With L fixed, the estimate had a percent difference of 3.7%. These are both extremely close to the sensor measurement, confirming the consistency of our experiment with the equation $F = ILB$. Possible sources of error include the volatility of the power supply and human error adjusting the scale.

Introduction

The force on a straight current-carrying wire in a magnetic field is given by $\vec{F} = I\vec{L} \times \vec{B}$, where I is the current, $\vec{L}$ is the length vector of the wire, and $\vec{B}$ is the magnetic field.

When the wire is perpendicular to the field, this simplifies to $F = ILB$. Using the right hand rule, we can determine that if the permanent magnet is placed with the “colorful” side facing up, palcing a current-carrying wire through the sides of the magnet opening will result in a vertical (upward/downward) force on the wire, called the *magnetic force*.

This magnetic force will cause an apparent change in mass when the wire is placed on a scale, which can be used to experimentally determine the magnetic field strength B.

Equipment

- Permanent Magnet from PASCO EM-8933 Kit

Parts 1 and 2:

- Manual Scale
- Current Loops (SF 37–42) from PASCO EM-8933 Kit
- Current-generating Power Supply
- Digital Multimeter (Ammeter)

Part 3:

- Magnetic Field Sensor

Procedure

1. **Part 1 (Force vs. Length):** With the current fixed at 1 A, each of the six current loops (with different wire lengths) was placed in the magnetic field created by the permanent magnets. The scale reading was recorded for each loop. The magnetic force was calculated from the change in apparent mass. A linear regression was performed over multiple trials to estimate the magnetic field inside the permanent magnet.

2. **Part 2 (Force vs. Current):** Using the SF 38 loop ($L = 4.2$ cm), the current was varied from 0.5 A to 5 A in 0.5 A increments. The current loops were placed in the magnetic field, and the scale readings were recorded. The magnetic force was calculated, and a linear regression was performed to estimate the magnetic field.

3. **Part 3 (Direct Measurement):** The magnetic field between the permanent magnets was measured directly using a magnetic field sensor, with three measurements taken.

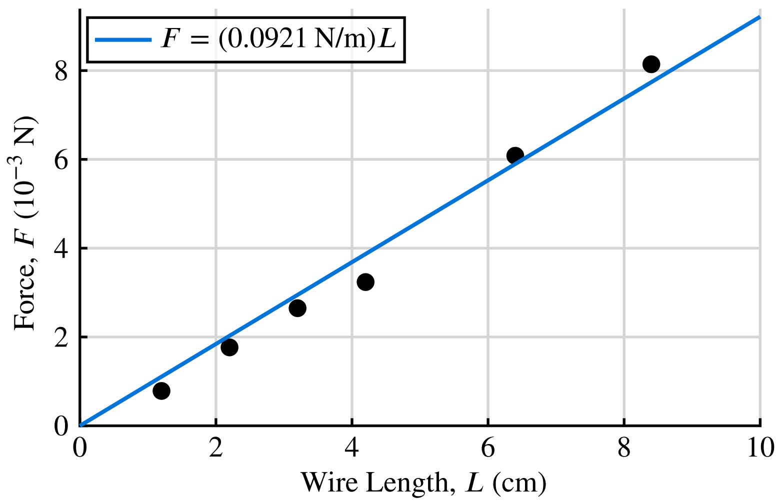
Part 1: Magnetic Force w/ Varying Wire Length

The current was fixed at $I = 1$ A. The base mass (no current) was $m_0 = 166.71$ g.

#	**Loop**	L	m	$\Delta m = m - m_0$	$F = (\Delta m)g$
1	SF 42	8.4 cm	167.54 g	0.83 g	$8.14 \cdot 10^{-3}$ N
2	SF 41	6.4 cm	167.33 g	0.62 g	$6.08 \cdot 10^{-3}$ N
3	SF 38	4.2 cm	167.04 g	0.33 g	$3.24 \cdot 10^{-3}$ N
4	SF 39	3.2 cm	166.98 g	0.27 g	$2.65 \cdot 10^{-3}$ N
5	SF 37	2.2 cm	166.89 g	0.18 g	$1.77 \cdot 10^{-3}$ N
6	SF 40	1.2 cm	166.79 g	0.080 g	$7.85 \cdot 10^{-4}$ N

Where L is the "effective" length of each wire affected by the magnetic field, m is the absolute mass measured on the manual balance, and Δm is the extra apparent mass due to the magnetic force.

Relative Magnetic Force with Varying Wire Length, fixed at 1 A of current



Performing a linear regression using Desmos, the slope of the best-fit line is 0.0921 N/m.

Since $F = ILB$, if F varies with L, then the slope equals IB:

$$\text{slope} = IB \quad \Rightarrow \quad B = \frac{\text{slope}}{I} = \frac{0.0921\text{ N/m}}{1\text{ A}} = 0.0921\text{ T}.$$

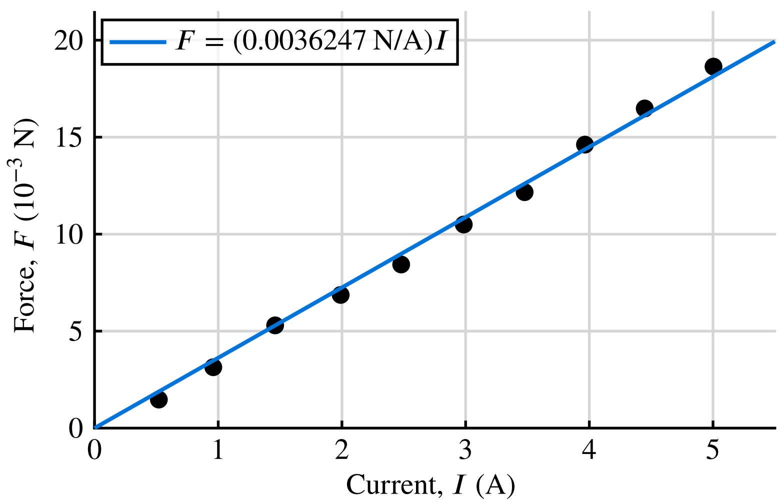
Part 2: Magnetic Force w/ Varying Current

The wire length was fixed at $L = 4.2$ cm (SF 38). The base mass was $m_0 = 166.71$ g.

#	I_{target}	I_{DMM}	m	$\Delta m = m - m_0$	$F = (\Delta m)g$
1	0.5 A	0.52 A	166.86 g	0.15 g	$1.47 \cdot 10^{-3}$ N
2	1.0 A	0.96 A	167.03 g	0.32 g	$3.14 \cdot 10^{-3}$ N
3	1.5 A	1.46 A	167.25 g	0.54 g	$5.30 \cdot 10^{-3}$ N
4	2.0 A	1.99 A	167.41 g	0.70 g	$6.87 \cdot 10^{-3}$ N
5	2.5 A	2.48 A	167.57 g	0.86 g	$8.44 \cdot 10^{-3}$ N
6	3.0 A	2.99 A	167.78 g	1.07 g	$1.05 \cdot 10^{-2}$ N
7	3.5 A	3.48 A	167.95 g	1.24 g	$1.22 \cdot 10^{-2}$ N
8	4.0 A	3.97 A	168.20 g	1.49 g	$1.46 \cdot 10^{-2}$ N
9	4.5 A	4.45 A	168.39 g	1.68 g	$1.65 \cdot 10^{-2}$ N
10	5.0 A	5.00 A	168.61 g	1.90 g	$1.86 \cdot 10^{-2}$ N

Where I_{target} was the current we tried to obtain on the power supply, I_{DMM} was the actual current measured by the DMM, m was the absolute mass measured on the balance, and Δm was the extra apparent mass due to the magnetic force.

Relative Magnetic Force with Varying Current, fixed at 4.2 cm **of wire length**



Performing a linear regression using Desmos, the slope of the best-fit line is 0.0036247 N/A.

Since $F = ILB$, if F varies with I, then the slope equals LB:

$$\text{slope} = LB \quad \Rightarrow \quad B = \frac{\text{slope}}{L} = \frac{0.0036247 \text{ N/A}}{0.042 \text{ m}} = 0.0863 \text{ T}.$$

Part 3: Direct Measurement of Magnetic Field

The magnetic field between the magnet was measured directly using a magnetic field sensor.

Trial	**Measured Magnetic Field *B***
1	88.70 mT
2	90.97 mT
3	88.87 mT
Average	89.51 mT

Comparison of Methods

Method	***B* (T)**	**% Difference**
Part 1 (Force vs. Length)	0.0921 T	2.9%
Part 2 (Force vs. Current)	0.0863 T	3.7%
Part 3 (Direct Measurement)	0.0895 T	—

All three methods yield consistent results. The percent difference between the indirect measurements (Parts 1 and 2) and the direct measurement (Part 3) is small.

The small variation between the methods can be attributed to several sources of uncertainty which are detailed in the *Error Analysis* below.

Error Analysis

Sources of uncertainty include:

- **human error from adjusting the scale**, since the scale was very sensitive to small vibrations. This would have caused random variations in the apparent mass m.
- **volatility of the current**, since the power supply had arbitrary fluctuations in current. This would have caused random variations in the measured current I.
- **human error adjusting the current of the power supply**, since the dial was extremely sensitive and it was rare that we are able to set the current to *exactly* the desired current. This would have caused systematic errors in the measured current I.
- **resistance from the power supply to the wire**, which would would lower the actual current in the wire. Since this "mystery" resistance should remain relatively constant, this is a systematic error which would have caused a consistent underestimation of the measured current I.

Conclusion

1. Find out the typical magnetic field in Tesla associated with:
   a. junkyard magnet (used to pick up cars): **1 T to 2 T**
   b. refrigerator magnet: **5 mT to 10 mT**
   c. Earth's magnetic field: **25 μT to 65 μT**

2. Assume that a square coil has 8 loops and 2 A of current: $N = 8$, $I = 2$ A.

   For each of the magnetic field strengths in Question 1, calculate the magnetic force on each side of the square coil if the field is parallel with the coil axis, given the side length of the square coil is (i) 1 cm and (ii) 10 cm.

   Let the side length of the square coil be L, and let $\hat{k}$ be the unit vector piercing through the coil. At all times, $\vec{L}$ is orthogonal to $\hat{k}$, so $\left|\vec{L} \times \hat{k}\right| = L$.

   a. junkyard magnet: let $\vec{B} = 1.5\text{ T}\,\hat{k}$.
      i. $L = 1$ cm: $F = NILB = 8 \cdot 2\text{ A} \cdot 0.01\text{ m} \cdot 1.5\text{ T} = 2.40 \cdot 10^{-1}\text{ N}$
      ii. $L = 10$ cm: $F = NILB = 8 \cdot 2\text{ A} \cdot 0.1\text{ m} \cdot 1.5\text{ T} = 2.40\text{ N}$

   b. refrigerator magnet: let $\vec{B} = 7.5\text{ mT}\,\hat{k} = 7.50 \cdot 10^{-3}\text{ T}\,\hat{k}$.
      i. $L = 1$ cm: $F = NILB = 8 \cdot 2\text{ A} \cdot 0.01\text{ m} \cdot 0.0075\text{ T} = 1.20 \cdot 10^{-3}\text{ N}$
      ii. $L = 10$ cm: $F = NILB = 8 \cdot 2\text{ A} \cdot 0.1\text{ m} \cdot 0.0075\text{ T} = 1.20 \cdot 10^{-2}\text{ N}$

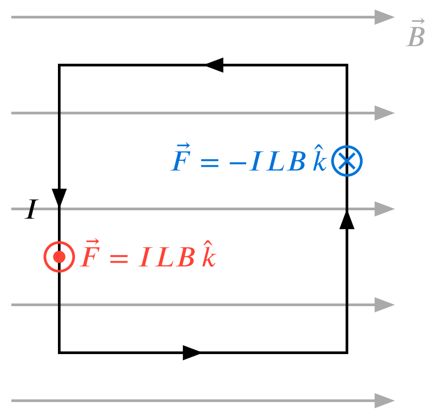
   c. Earth's magnetic field: let $\vec{B} = 45\text{ μT}\,\hat{k} = 4.50 \cdot 10^{-5}\text{ T}\,\hat{k}$.
      i. $L = 1$ cm: $F = NILB = 8 \cdot 2\text{ A} \cdot 0.01\text{ m} \cdot 4.5 \cdot 10^{-5}\text{ T} = 7.20 \cdot 10^{-6}\text{ N}$
      ii. $L = 10$ cm: $F = NILB = 8 \cdot 2\text{ A} \cdot 0.1\text{ m} \cdot 4.5 \cdot 10^{-5}\text{ T} = 7.20 \cdot 10^{-5}\text{ N}$

Table of Calculations

Magnet Type	**F ($L = 1$ cm)**	**F ($L = 10$ cm)**
Junkyard (1.5 T)	$2.40 \cdot 10^{-1}$ N	2.40 N
Refrigerator (7.5 mT)	$1.20 \cdot 10^{-3}$ N	$1.20 \cdot 10^{-2}$ N
Earth (45 μT)	$7.20 \cdot 10^{-6}$ N	$7.20 \cdot 10^{-5}$ N

3. For the configuration $B = 1$ T, $L = 10$ cm, $N = 8$, and $I = 2$ A, with $\vec{B}$ perpendicular to the coil axis (i.e. in the plane of the coil), consider the four sides of the square coil.

   a. Draw the forces acting on each side of the coil.



The magnetic force on the top and bottom sides are omitted because they are zero.
The gravitational force is omitted because it is negligible.

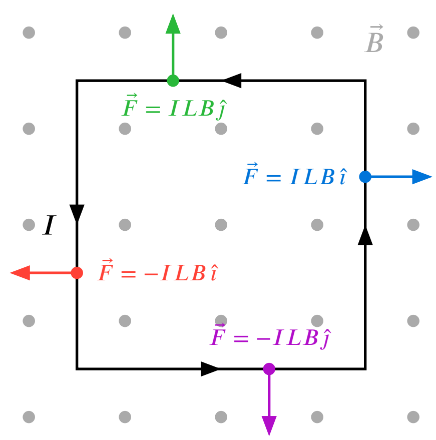
   b. What is the net force on the coil?

   The two magnetic forces acting on the left and right sides of the coil in the diagram cancel out, so there is zero net force on the coil.

   However, there is a net *torque* that would rotate the coil since the magnetic forces act in opposite directions, and the points at which the forces are exerted are away from the coil's center of mass. (The torque would tend to rotate the coil counterclockwise.)

4. What happens if the magnetic field is parallel to the axis of the coil?

   The net force on the wire is still zero:



   As observed, both the horizontal and vertical magnetic forces cancel on opposite sides of the coil. However, the difference from (q3) is that now, there is no more net torque on the coil. The coil is in rotational equilibrium as well.

5. Let $\vec{\mu}$ be the *magnetic dipole moment* of the loop.

   a. What is the definition of the magnetic dipole moment $\vec{\mu}$ of the loop?

   Mathematically, the magnetic dipole moment of a current loop is defined as:

   $$\vec{\mu} = N I A \hat{k}.$$

   The magnitude μ is the ideal torque that can be exerted on the dipole per unit magnetic field strength. That is, if the dipole is placed in a magnetic field $\boldsymbol{B}$, the maximum torque that can be exerted on the dipole is $\mu \boldsymbol{B}$.

   The direction $\hat{k}$ is given by the right hand rule with respect to the current flow.

   b. Calculate the magnetic moment for these loops ($\boldsymbol{L} = 1$ cm and $\boldsymbol{L} = 10$ cm).

      i. When $L = 1$ cm, $\vec{\mu} = N I A\hat{k} = N I L^2\hat{k} = 8 \cdot 2\text{ A} \cdot (1\text{ cm})^2\hat{k} = 16\text{ A cm}^2\,\hat{k}$.

      ii. When $L = 10$ cm, $\vec{\mu} = N I A\hat{k} = N I L^2\hat{k} = 8 \cdot 2\text{ A} \cdot (10\text{ cm})^2\hat{k} = 1600\text{ A cm}^2\,\hat{k}$.

6. Find the maximum magnitude of the torque that the coil can experience.

   The torque is given by $\vec{\tau} = \vec{\mu} \times \vec{B}$, which means $\left|\vec{\tau}\right| = \mu B \sin\theta$, where θ is the angle between $\vec{\mu}$ and $\vec{B}$. The maximum magnitude of the torque occurs when $\sin\theta = 1$, i.e. when $\theta = 90°$. Thus, the maximum magnitude of the torque is given by $\mu \boldsymbol{B}$.

   For the value of $\mu = 16\text{ A cm}^2$ and $\boldsymbol{B} = 1$ T, the maximum torque is:

   $$\tau_{\text{max}} = \mu \boldsymbol{B} = 16\text{ A cm}^2 \cdot 1\text{ T} = 16\text{ N m}.$$

   For the value of $\mu = 1600\text{ A cm}^2$ and $\boldsymbol{B} = 1$ T, the maximum torque is:

   $$\tau_{\text{max}} = \mu \boldsymbol{B} = 1600\text{ A cm}^2 \cdot 1\text{ T} = 1600\text{ N m}.$$